Group 26

Pet Minder

ECS506U Software Engineering Group Project

Requirements Elicitation Report

1. List of Requirements

Functional Requirements

ID	Requirement	Type	Priority	Use Case
1	The system shall allow Pet Owners and Pet Minders to create a profile including their name, location, and contact details.	Functional (D)	Core	Create Profile
3	The system shall allow Pet Minders to add "animal tags" on their profile to indicate types of pets they are comfortable minding.	Functional (D)	Core	Create Profile
4	The app should allow Pet Owners to register a pet profile by specifying the animal type (Dog/Cat) and breed.	Functional (D)	Core	Add Pet
5	The system shall allow Pet Owners to request a veterinary professional to upload verified medical records to a pet's profile.	Functional (D)	Core	Add Pet Info
6	The system shall provide a search interface for Pet Owners to filter minders by location, price range, and rating.	Functional	Core	Search for Minder
7	The system shall provide a search interface for Pet Minders to filter jobs by location, time, job type, and animal type.	Functional	Core	Search for Job
8	The system shall allow Pet Owners to send a booking request to a Pet Minder for a specific date and time window.	Functional	Core	Request Booking
9	The system shall allow Pet Minders to accept a booking invitation,triggering an automated confirmation to the Pet Owner.	Functional	Core	Accept Job
10	The system shall provide Pet Minders with an interface to view pending booking invitations, dates, pet types, and owner instructions.	Functional	Core	View Job Requests
11	The system shall allow Pet Minders to decline a booking invitation if the timing or pet requirements do not align with their current circumstances.	Functional	Core	Decline Job
12	The system shall allow Pet Minders to set their status to "unavailable" for specific date ranges, preventing new booking requests during those times.	Functional	Core	Manage Availability
13	The system shall facilitate an in-app chat for users to communicate regarding booking details.	Functional	Core	Request Booking, Chat
14	The system shall provide a real-time GPS tracking interface for Pet Owners to monitor their pet’s location during a scheduled walk.	Functional	Core	Track Minder
15	The system shall allow Pet Owners to edit and update pet information after the pet profile has been created.	Functional (D)	Core	Add Pet Info
16	The system shall allow Pet Owners to view available Pet Minders through a profile browsing interface.	Functional	Core	Browse Profiles

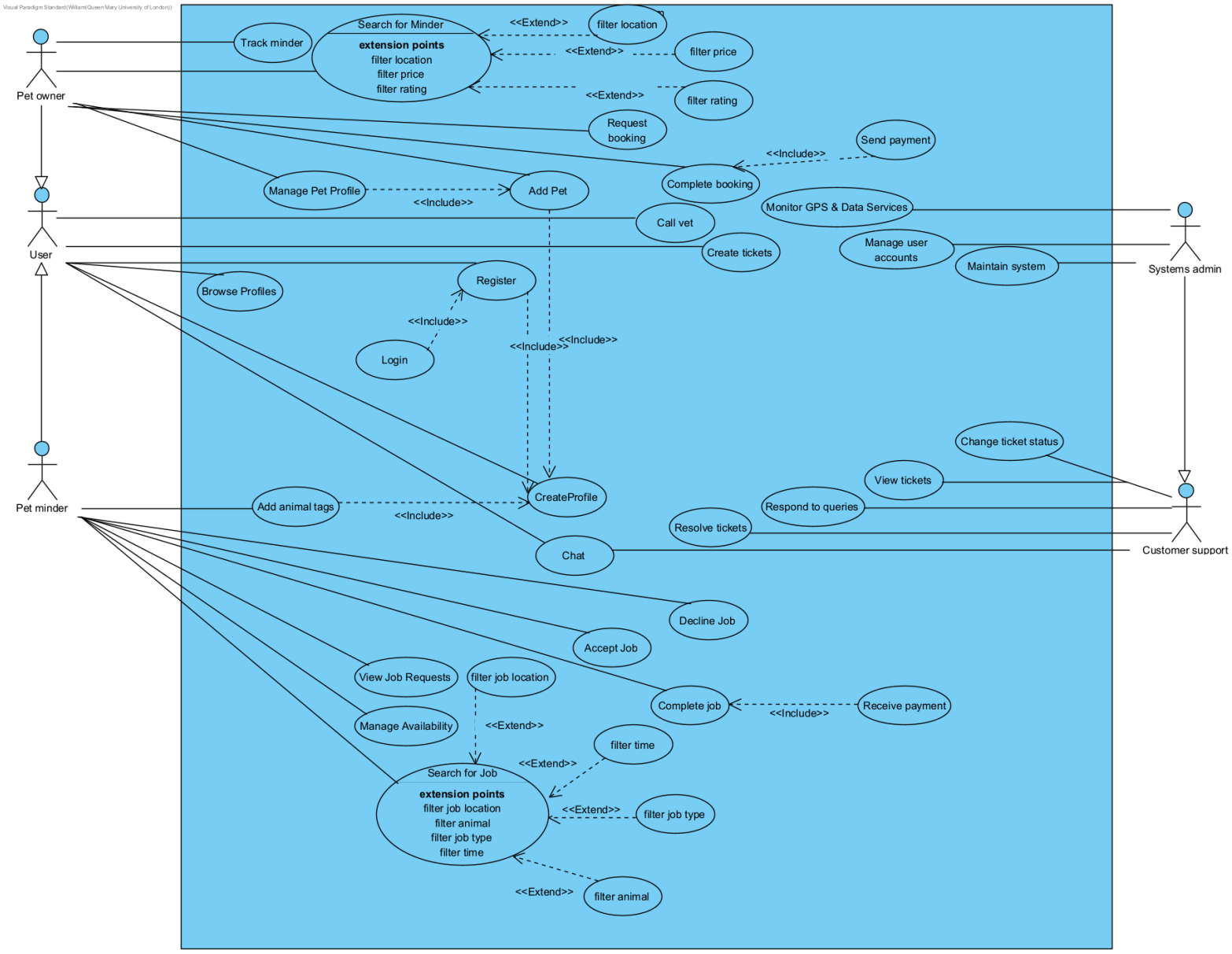
17	The system shall recommend Pet Minders to Pet Owners based on their past bookings and preferences.	Functional	Optional	Browse Profiles
18	The system shall allow Pet Owners to select a preferred date and time when booking a Pet Minder.	Functional	Core	Choose time & date
19	The system shall notify Pet Minders when a new booking request is received.	Functional	Core	Request Booking
20	The system shall allow Pet Minders to update their availability status for accepting new jobs.	Functional	Core	Search for Job
21	The system shall allow Pet Owners to cancel a pending booking request before it has been accepted by a Pet Minder.	Functional	Core	Book Minder
22	The system shall notify a Pet Owner if their booking request is declined by a Pet Minder.	Functional	Core	Decline Job
23	The system shall allow Pet Minders to set a pricing rate on their profile that is visible to Pet Owners.	Functional	Optional	Create Profile
24	The system shall allow Pet Owners to view booking details including time, location, and assigned Pet Minder.	Functional	Core	Request Booking
25	The system shall allow users to securely log in before accessing system features.	Functional	Core	Login
26	The system shall allow new users to register an account using personal details.	Functional (D)	Core	Register
27	The system shall allow Pet Minders to manage and update their profile information including location and contact details.	Functional (D)	Core	Create Profile
28	The system shall allow each user to be identified with a unique username.	Functional (D)	Core	Create Profile
29	The system shall require Pet Minders and Pet Owners to choose and enter a method of preferred communication.	Functional (D)	Core	Create Profile
30	The system shall allow Pet Owners to add multiple pets to their profile/account.	Functional (D)	Core	Add Pet
31	The system shall allow Pet Minders and Owners to review the other party upon the end of a minding session.	Functional	Core	Review Profile
32	The system shall allow all users to add a preferred veterinary clinic's contact details to their profile.	Functional	Core	Create Profile
33	The system shall allow all users to view job requests posted by the opposing user type.	Functional	Core	View Job Request
34	The system shall allow Pet Owners to view all requests from Pet Minders for a specific listing.	Functional	Core	Request Booking
35	The system shall allow all users to post listings.	Functional	Core	Add Job Request
36	The system shall allow all users to add a geographical area or location to their listings.	Functional	Core	Add Job Request
37	The system shall allow Pet Minders and Pet Owners to make queries or complaints in the form of a ticket.	Functional	Core	Create Ticket

38	The system shall allow Administrators to view tickets submitted by users.	Functional	Core	View Tickets
39	The system shall allow Administrators to sort tickets by ascending or descending priority, and grouped categories.	Functional	Core	View Tickets
40	The system shall allow Customer Support to view all tickets that are not categorised as a technical issue.	Functional	Core	View Tickets
41	The system shall allow Administrators to use the chat function to converse with the user associated with an active ticket.	Functional	Core	Resolve Ticket
42	The system shall allow tickets to have a status, and to be assigned the status Pending automatically upon submission.	Functional	Core	Create Ticket
43	The system shall allow a ticket's status to be updated to Opened upon an Administrator confirming they will resolve the ticket.	Functional	Core	Resolve Ticket
44	The system shall allow Administrators to update a ticket's status to Closed once the ticket is resolved.	Functional	Core	Resolve Ticket
45	The system shall allow users to reset their password via a verified email link if they have forgotten it.	Functional	Core	Login
46	The system shall allow users to permanently delete their account and all associated personal data in compliance with UK GDPR.	Functional (D)	Core	Create Profile
47	The system shall allow Administrators to suspend or remove a user account in response to a reported misconduct ticket.	Functional	Core	Resolve Ticket
48	The system shall allow Pet Owners to save Pet Minders to a favourites list for quick future access.	Functional	Optional	Browse Profiles
49	The system shall allow Pet Owners to save their selected search filter preferences for use in future searches.	Functional	Optional	Search For Minder
50	The system shall allow Pet Owners to make secure in-app payments to Pet Minders upon booking confirmation.	Functional	Core	Make payment
51	The system shall allow Pet Minders to upload certifications or proof of experience to their profile.	Functional	Optional	Create Profile
52	The system shall allow Pet Owners to set up recurring bookings with the same Pet Minder on a weekly or custom schedule.	Functional	Optional	Request Booking

Non-Functional Requirements

ID	Requirement	Type	Priority	Use Case
1	All user passwords will be hashed using industry standard encryption (e.g. bcrypt, Argon2 or supabase auth) and should not be stored or transmitted in plain text.	Non-functional (Security - Access Control)	Core	Login, Register
2	The system shall enforce HTTPS (TLS 1.2 or higher) encryption for all data transmitted between client devices and the server.	Non-functional (Security - Data Transmission)	Core	All Use Cases
3	GPS tracking updates during an active pet walking session should refresh at least every 10 seconds while the session is active.	Non-functional (Performance - Real Time Tracking)	Core	Track Pet Session
4	The mobile application should support background GPS tracking on both IOS and Android.	Non-functional (Platform/ Reliability)	Core	Track Pet Session
5	The system should achieve 99% uptime availability measured monthly, excluding scheduled maintenance.	Non-functional (Availability)	Core	All Use Cases
6	The system should respond to user actions (e.g. opening profiles, sending messages, booking requests) within 5 seconds minimum under normal network conditions.	Non-functional (Performance - Response Time)	Core	Browse Profiles, Chat, Booking
7	User data, including medical records and booking history should be stored securely and comply with UK GDPR data protection regulations.	Non-functional (Security - Data Protection/ Legal Compliance)	Core	Manage Pet Profile
8	The system should support concurrent usage of at least 1000 active users without performance degradation.	Non-functional (Scalability)	Optional	All Use Cases
9	The application interface should follow mobile usability standards and ensure that core tasks (booking, messaging, viewing, tracking) can be completed.	Non-functional (Usability)	Core	Booking, Chat
10	The system should automatically back up all booking, user, and pet data 24/7 to prevent data loss in case of system failure.	Non-functional (Reliability - Data Recovery)	Core	All Use Cases

2. Use-Case Diagram



3. Use Case Description

Use Case - Search For Minder

This use case depicts a Pet Owner looking for a Pet Minder using the search interface.

Actors: Pet Owner

Preconditions: The Pet Owner must have registered and logged in to their account. The system must also be online and functional.

Basic Flow:

1. The pet owner selects the interface 'Search for Minder'
2. The system displays the interface and shows unfiltered results of Pet Minders.
3. The Pet Owner can select any filter (location, price or rating) to narrow their search
4. If any filter is chosen, the system must retrieve all pet minders that meet the filters.
5. The system updates the display with the pet minders retrieved.

Alternate Flows:

1. Location filter applied. The system retrieves matches that match with any other filter currently applied and updates display.
2. Price filter applied. The system retrieves matches that match with any other filter currently applied and updates display.
3. Rating filter applied. The system retrieves matches that match with any other filter currently applied and updates display.
4. No filter applied. The system does not update the display, showing unfiltered results.

Postconditions:

- Pet Owner is shown the filtered/unfiltered pet minders on display and is able to view their profiles.

Use case - Request booking

A Pet Owner can request a booking with a selected Pet Minder for a specific date and time.

Actors: Pet Owner

Preconditions:

1. The Pet Owner is logged in and the system is online.
2. The Pet Owner has used the Search for Minder interface and is shown the final results of Pet Minders.

3. The Pet Owner has chosen one of the Pet Minders profile to request a booking.

Basic Flow:

1. Pet Owner chooses the Booking interface.
2. The system shows an interface showing times and days when the Pet Minder is available.
3. Pet Owner selects their preferred date and time and confirms booking.
4. The system notifies the Pet Minder of the booking.

Alternate Flow:

1. Pet Owner selects a Pet Minder that is not unavailable for their preferred date and time and must choose another.
2. If after booking and the Pet Owner has felt that they suddenly can't do the chosen time, the Pet Owner can cancel the pending request.

Postconditions:

- A pending booking request is in the system and is just waiting to be accepted/declined by the Pet Minder.
- The Pet Minder has been notified of the booking.
- The Pet Owner can see their booking details.

4. Risk Assessment

Risk	Likelihood	Severity	Impact	Preventative / Mitigating Actions
Requirements are incomplete	Low	Medium	Unfinished product, low-quality product.	Define requirements and review their progress regularly. Ensure members are satisfied with the level with which each requirement is completed.
Members are not doing the work allocated to them	Medium	High	Unfinished product, unhappy members, tension, low-quality product.	Make sure constant meetings are in place - ensuring that we can have micro deadlines and discussions of other teammates' contributions
Scope creep	Medium	Medium	Continuous changes in scope result in delay in delivery and incomplete essential functions.	Set clear goals and objectives for the project plan. Prioritise these objectives and ensure scope does not grow during progression on the project.
Low team motivation	Medium	High	Low-quality product/ Unfinished product.	If there is a root cause for the lack of motivation, members have a meeting to make sure everyone is happy. Otherwise, other team members should motivate the unmotivated member.
Delays in submission due to procrastination or sickness	Medium	High	Low-quality product/ Stress to the rest of the team/ Penalty to marks.	Make sure there is constant communication among teammates so everyone stays organised and

				completes their work on time.
Uneven workloads	High	High	Overworked members/ Inequality in responsibility.	Make sure that before each section, members divide the section equally among themselves, with everyone agreeing on their parts.
Inconsistent quality in work	Medium	Medium	Underdeveloped sections in project/ Low-quality product.	Have frequent discussions and peer review every person's work to make sure everyone is happy with the entire product.
Sources and data are incorrect or outdated	Low	Medium	Inefficient product from inaccurate information.	Make sure every teammate correctly finds information from proper sources. Have another member peer review sources of reference, if needed.
Delays due to lack of required infrastructure (Software, Hardware)	Low	Medium	Inability to do certain sections of the product (use-case diagrams and code) delaying progression.	Have every member download the needed software - ensure they have updated versions to match.
Requirements are ambiguous	Low	Low	Members put requirements that can't be integrated efficiently or create a misconfigured product to what is intended.	Make sure members discuss what a feasible requirement is that does not hinder full development of the product.
Client has inaccurate expectations of product	Low	High	Client dissatisfaction	Clear project plans and expectations are discussed during client meetings and

				presentations, with every member agreeing on all of their requirements before presenting them.